Evolutionary Microbiology Group

Prof. Dr. Tal Dagan
Institute of General Microbiology, Faculty of Mathematics and Natural Sciences

CAU
Christian-Albrechts-Universität zu Kiel
Mathematisch-Naturwissenschaftliche Fakultät

Bacteria are found everywhere. Unlike eukaryotes, bacterial evolution comprises both vertical and horizontal components. Recombination at the species level plays a role in selective sweeps through the population, while inter-species lateral gene transfer (LGT) has important implications to microbial adaptation and evolutionary transformations. In our research we study molecular and genome evolution of microbial genomes.

Experimental Section

Experimental evolution
Genetic engineering
Natural transformation

Teaching of:
biol254 Evolution and biology of lateral gene transfer

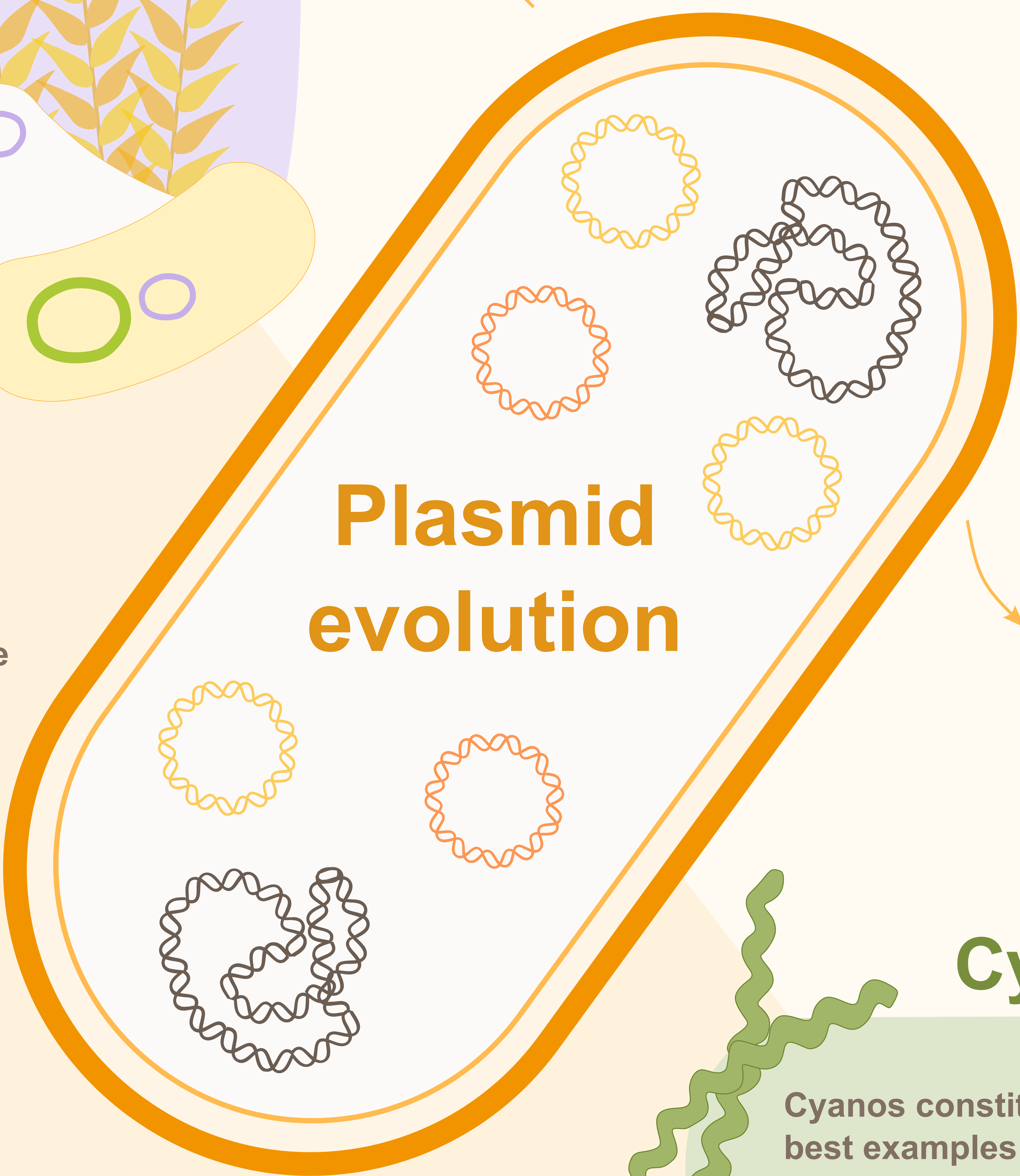
Symbiosis

Bacteria associated with eukaryotic hosts can have effects on the host fitness and biology. In turn host colonization dynamics may impact bacterial genome evolution.
Projects:

- **Colonization dynamics in the wheat microbiota (e.g. *Pantoea agglomerans*)**
- **Study plasmid encoded host-associated 'lifestyle' genes**
- **Population genetics and genome evolution in symbiosis**

Plasmid persistence remains an evolutionary paradox. As they can be costly to cells but still can persist over evolutionary timescales without selection for the plasmid function.
Projects:

- **Emergence of plasmid stability**
- **Evaluation of the evolutionary success of plasmid genotypes within the host**



Plasmid evolution

Plasmids are present in multiple copies, leading to intracellular genetic heterogeneity. Plasmid diversity is subject to changes upon cell division due to the random segregation of plasmid copies and the contribution of novel alleles via lateral gene transfer.
Projects:

- **Effect of intracellular genetic drift**
- **Contribution of lateral gene transfer to plasmid allele dynamics**

Plasmids accesory genes can encode diverse fuctions that could lead to bacterial host adaptation or innovation. Therefore, traits encoded on plasmids might follow the rules and factors underlying plasmid genome evolution.
Projects:

- **Prevalence and contribution of duplications and pseudogenization in plasmid genome evolution**
- **Population genetics modelling of the fixation dynamics of multicopy alleles**
- **Plasmid genome evolution with a focus on the accessory gene distribution (e.g. antimicrobial resistance gene)**

Cyanobacteria

Cyanos constitute a diverse phylum that include the best examples of prokaryotic multicellularity. Furthermore, their capacity for natural transformation has implications for their genome evolution and is very sought after for biotechnological applications.
Projects:

- **Experimental verification of naturally transformable species**
- **Phylogenetic reconstruction of ecological and phenotypic trait evolution (e.g. natural competence and multicellularity)**

Phylogenetic trees play a cardinal role in evolutionary theory as they are used to describe and investigate evolutionary relations between entities.
Projects:

- **Development of a novel phylogenetic rooting method using minimal ancestor deviation**

Phylogenetic methods

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Comparative genomics
Computational modelling
Phylogenomics

Computational Section

Teaching of:
biol109 Biostatistics
biol124 Bioinformatics
biol160 Molecular evolution
biol258 Comparative genomics